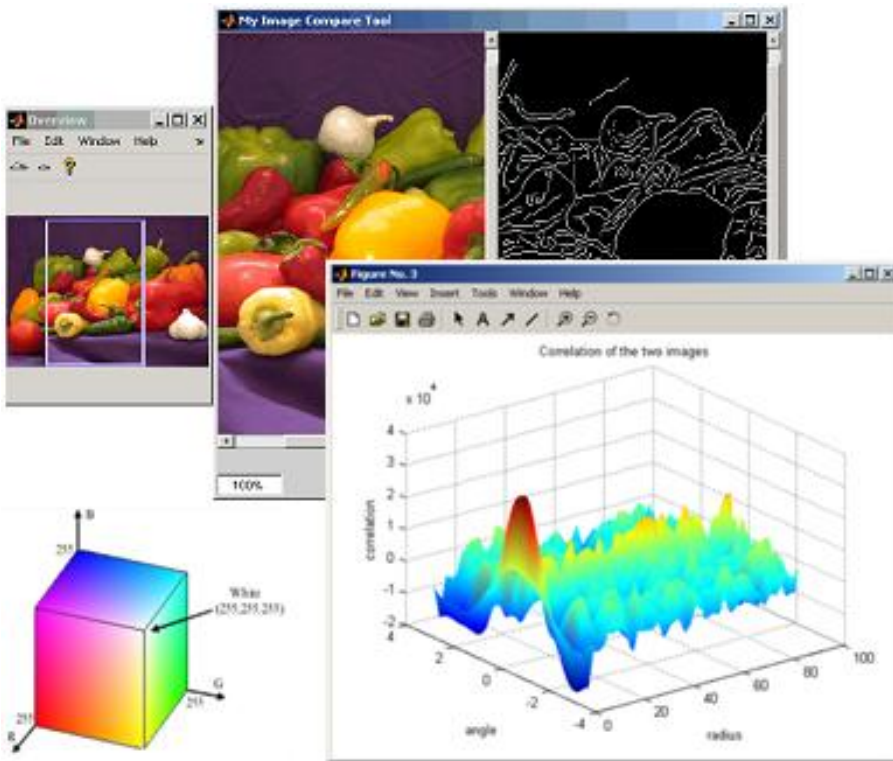


Digital Image Processing

LECTURE 6: PIXEL RELATIONSHIP



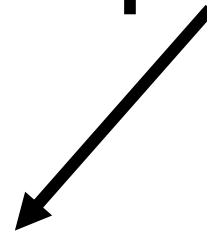
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Relationship Between Pixels

Relationship Between Pixels

Image

Topology



the way in which constituent parts
are interrelated or arranged.

Relationship Between Pixels

□ Image Topology:

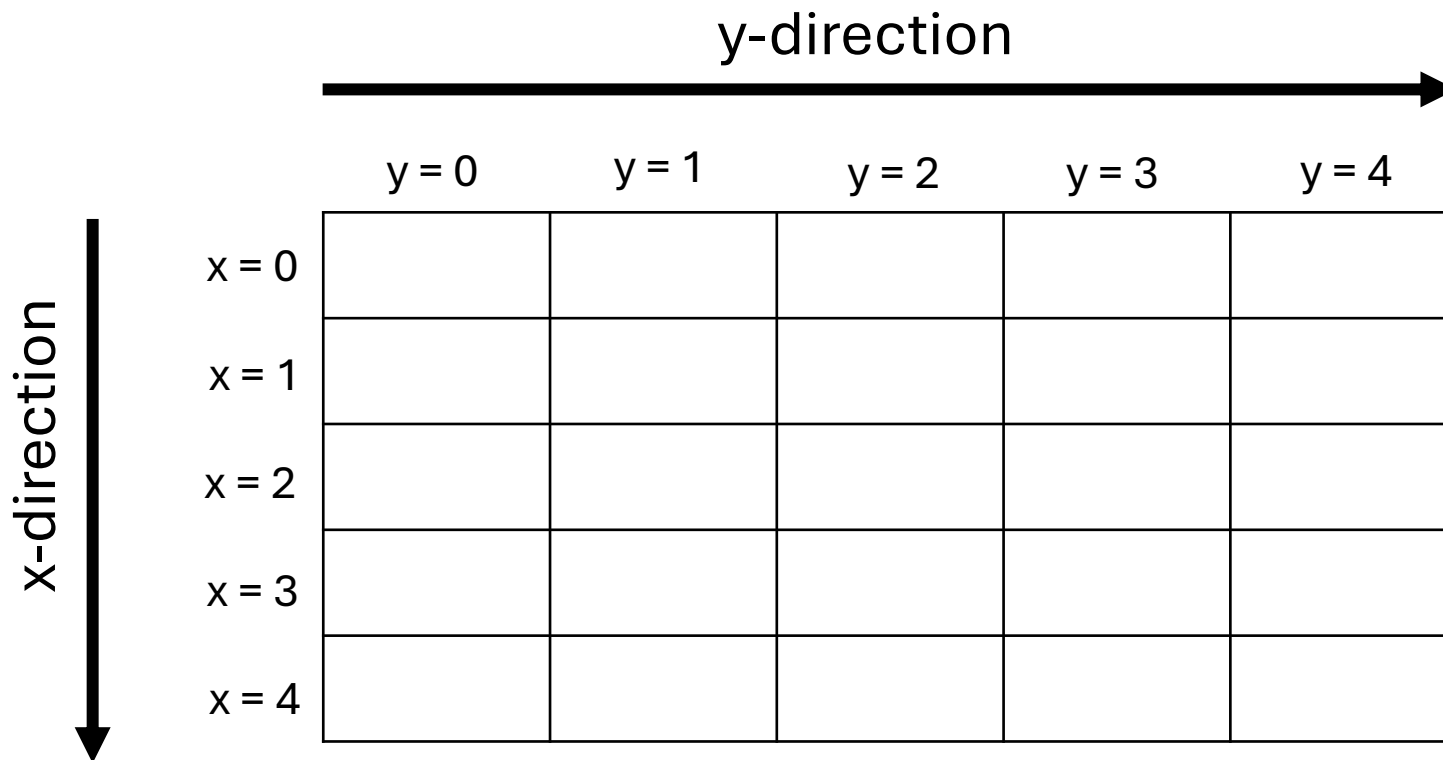
- Image topology is a branch of image processing that deals with the fundamental properties of the image such as image neighbourhood, paths among pixels, boundary and connected components.
- It characterizes the image with topological properties such as neighbourhood, adjacency and connectivity.

□ Different Fundamental Properties:

- Neighbourhood relationship between pixels
- Connectivity
- Adjacency
- Path
- Connected components
- Region and Boundaries

Basic Relationship between Pixels

- An image is denoted by a function $f(x, y)$
- Each element $f(x, y)$ at location (x, y) is called a pixel
- There exist some basic and important relationships between pixels



Basic Relationship between Pixels

- An image is denoted by a function $f(x, y)$
- Each element $f(x, y)$ at location (x, y) is called a pixel
- There exist some basic and important relationships between pixels

y-direction 

	y = 0	y = 1	y = 2	y = 3	y = 4
x = 0	$f(0,0)$	$f(0,1)$	$f(0,2)$	$f(0,3)$	$f(0,4)$
x = 1	$f(1,0)$	$f(1,1)$	$f(1,2)$	$f(1,3)$	$f(1,4)$
x = 2	$f(2,0)$	$f(2,1)$	$f(2,2)$	$f(2,3)$	$f(2,4)$
x = 3	$f(3,0)$	$f(3,1)$	$f(3,2)$	$f(3,3)$	$f(3,4)$
x = 4	$f(4,0)$	$f(4,1)$	$f(4,2)$	$f(4,3)$	$f(4,4)$

x-direction 

Discrete Image Representation

Discrete image representation in Cartesian coordinate system

	$y = 0$	$y = 1$	$y = 2$
$x = 0$	$f(0,0)$	$f(0,1)$	$f(0,2)$
$x = 1$	$f(1,0)$	$f(1,1)$	$f(1,2)$
$x = 2$	$f(2,0)$	$f(2,1)$	$f(2,2)$

Image Coordinates in
MATLAB

	$y = 1$	$y = 2$	$y = 3$
$x = 1$	$f(1,1)$	$f(1,2)$	$f(1,3)$
$x = 2$	$f(2,1)$	$f(2,2)$	$f(2,3)$
$x = 3$	$f(3,1)$	$f(3,2)$	$f(3,3)$

Neighbourhood Property

Neighbourhood Relationship

□ 4-Neighbours of p:

	$(x-1, y)$	
$(x, y-1)$	$P(x, y)$	$(x, y+1)$
	$(x+1, y)$	

■ Pixel corresponds to the location:

- $(x, y+1)$, and $(x, y-1)$ are Horizontal neighbours
- $(x-1, y)$, and $(x+1, y)$ are Vertical neighbours
- $(x, y-1)$, $(x, y+1)$, $(x-1, y)$ and $(x+1, y)$ are 4-neighbors of P and denoted as $N_4(p)$
- Each of these neighbours is at a unit distance from P.
- If P is a boundary pixel, then it will have less number of neighbours

Neighbourhood Relationship

□ Diagonal Neighbours of P:

$(x-1, y-1)$		$(x-1, y+1)$
	$P(x, y)$	
$(x+1, y-1)$		$(x+1, y+1)$

- Pixel corresponds to the location:
 - $(x-1, y-1)$, $(x-1, y+1)$, $(x+1, y-1)$ and $(x+1, y+1)$ are diagonal neighbours and denoted at $N_D(p)$.

Neighbourhood Relationship

□ 8-Neighbours of P:

$(x-1, y-1)$	$(x-1, y)$	$(x-1, y+1)$
$(x, y-1)$	$P(x, y)$	$(x, y+1)$
$(x+1, y-1)$	$(x+1, y)$	$(x+1, y+1)$

■ Pixel corresponds to the location:

- $(x, y-1)$, $(x, y+1)$, $(x-1, y)$ and $(x+1, y)$ are 4-neighbors of P and denoted as $N_4(p)$
- $(x-1, y-1)$, $(x-1, y+1)$, $(x+1, y-1)$ and $(x+1, y+1)$ are diagonal neighbours and denoted as $N_D(p)$.
- $(x, y-1)$, $(x, y+1)$, $(x-1, y)$, $(x+1, y)$, $(x-1, y-1)$, $(x-1, y+1)$, $(x+1, y-1)$ and $(x+1, y+1)$ are 8-neighbors of P and denoted as $N_8(p)$.

Neighbourhood Relationship

□ 8-Neighbours of P:

$(x-1, y-1)$	$(x-1, y)$	$(x-1, y+1)$
$(x, y-1)$	$P(x, y)$	$(x, y+1)$
$(x+1, y-1)$	$(x+1, y)$	$(x+1, y+1)$

▪ Pixel corresponds to the location:

- The points $N_4(p)$ and $N_D(p)$ together are called 8-neighbors of p .

$$N_8(p) = N_4(p) \cup N_D(p)$$

- Again, if P is a boundary pixel, then both $N_D(p)$ and $N_8(p)$ will have lesser number of pixel i.e.
 - $N_D(p)$ will be less than 4
 - $N_8(p)$ will be less than 8



Adjacency

Adjacency

Two pixels p and q are adjacent if they are connected:

- **4-adjacency:** Two pixels p and q with values from V are 4-adjacent if q is in the set $N_4(p)$.

$$V = \{0, 1\}$$

1	1	0
1	1 (p)	0
1	0	1

Adjacency

Two pixels p and q are adjacent if they are connected:

- **8-adjacency:** Two pixels p and q with values from V are 8-adjacent if q is in the set $N_8(p)$.

$$V = \{1, 2\}$$

0	1	1
0	2 (p)	0
0	0	1

Adjacency

Two pixels p and q are adjacent if they are connected:

- **m-adjacency:** Two pixels p and q with values from V are m-adjacent if
 - q is in $N_4(p)$, or
 - q is in $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose values are from V .

0	1	1
0	1	0
0	0	1



Connectivity

Connectivity

- Let S represent a subset of pixels in an image
- Two pixels p with coordinates (x_0, y_0) and q with coordinates (x_n, y_n) are said to be connected in S if there exists a path

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$$

PATH

- A path from pixel p with coordinates (x_0, y_0) to pixel q with coordinates (x_n, y_n) is a sequence of distinct pixels with coordinates $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$
where (x_i, y_i) and (x_{i-1}, y_{i-1}) are adjacent for $1 \leq i \leq n$
- Here n is the length of the path
- If $(x_0, y_0) = (x_n, y_n)$, the path is closed path
- We can define 4-, 8-, and m -paths based on the type of adjacency used

Numerical Examples

Ques1: Compute the length of shortest-4 path between pixels p and q, where $V = \{1, 2\}$

4	2	3	2 (q)
3	3	1	3
2	3	2	2
2 (p)	1	2	3

Numerical Examples

Ques2: Compute the length of shortest-8 path between pixels p and q, where $V = \{1, 2\}$

4	2	3	2 (q)
3	3	1	3
2	3	2	2
2 (p)	1	2	3

Numerical Examples

Ques3: Compute the length of shortest-m path between pixels p and q, where $V = \{1, 2\}$

4	2	3	2 (q)
3	3	1	3
2	3	2	2
2 (p)	1	2	3

Numerical Examples

Ques4: Consider the image segment as shown. Let $V = \{0, 1\}$, compute the length of the shortest 4-path, 8-path and m-path between the pixels p and q. If a particular path does not exist between these two points, Explain why?

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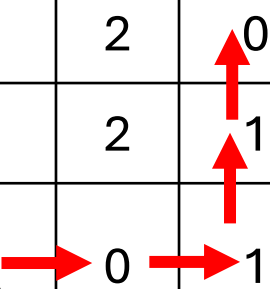
3	1	2	1(q)
2	2	0	2
1	2	1	1
1(p)	0	1	2

Numerical Examples

$$V = \{0, 1\}$$

shortest 4-path

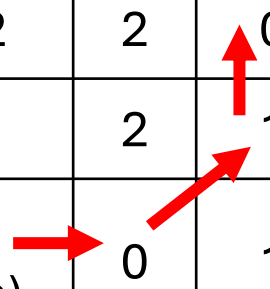
3	1	2	1 (q)
2	2	0	2
1	2	1	1
1 (p)	0	1	2



No Path

shortest 8-path

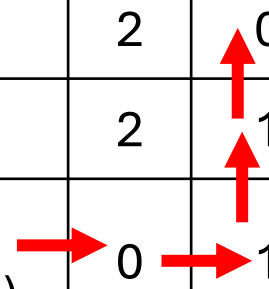
3	1	2	1 (q)
2	2	0	2
1	2	1	1
1 (p)	0	1	2



Length = 4

shortest m-path

3	1	2	1 (q)
2	2	0	2
1	2	1	1
1 (p)	0	1	2



Length = 5

Numerical Examples

Ques5: Consider the image segment as shown in Fig. 4. Let $V = \{0, 1\}$. Compute the length of the shortest 4- and 8-path between p and q .

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3	1	2	1(q)
2	2	0	2
1	2	1	1
1(p)	0	1	2

Fig. 4

Numerical Examples

Ques6: Consider the following image with marked pixels p and q, If $V = \{1, 2\}$; find the shortest 4-path, 8-path and m-path between pixels p and q

3	1	2	1(q)
2	2	0	2
1	2	1	1
1(p)	0	1	2

Numerical Examples

$$V = \{1, 2\}$$

shortest 4-path

3	1	2	1 (q)
2	2	0	2
1	2	1	1
1 (p)	0	1	2

shortest 8-path

3	1	2	1 (q)
2	2	0	2
1	2	1	1
1 (p)	0	1	2

shortest m-path

3	1	2	1 (q)
2	2	0	2
1	2	1	1
1 (p)	0	1	2

Numerical Examples

Ques7: Consider the image shown below. Find the shortest m-adjacent path from pixel p to q. $V = \{1, 2\}$. Draw the image in your answer sheet and mark the path.

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3	1	2	1(q)
2	2	0	2
1	2	1	1
1(p)	0	1	2

Numerical Examples

Ques 8: Consider the following 1-bit image

P: 1	1	1	1
1	1	0	1
0	0	0	1
0	0	1	0
0	0	1	Q:1

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- Draw the m-adjacency path from Pixel P to Pixel Q where $V = \{1\}$
- Find the size of the image.

Numerical Examples

Ques9: Consider the following 1-bit image

P: 1	1	1	0	0
1	1	0	0	1
1	0	1	1	1
1	1	1	0	Q:1
1	0	1	0	0

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- Draw the m-adjacency path from Pixel P to Pixel Q where $V = \{1\}$
- What will be the size of the given image if gray level range is [0 to 15]

Numerical Examples

Ques10: Consider the following image. Find the 4-adjacent and m-adjacent neighbours of the pixel P(1,1). $V = \{1, 2, 3\}$

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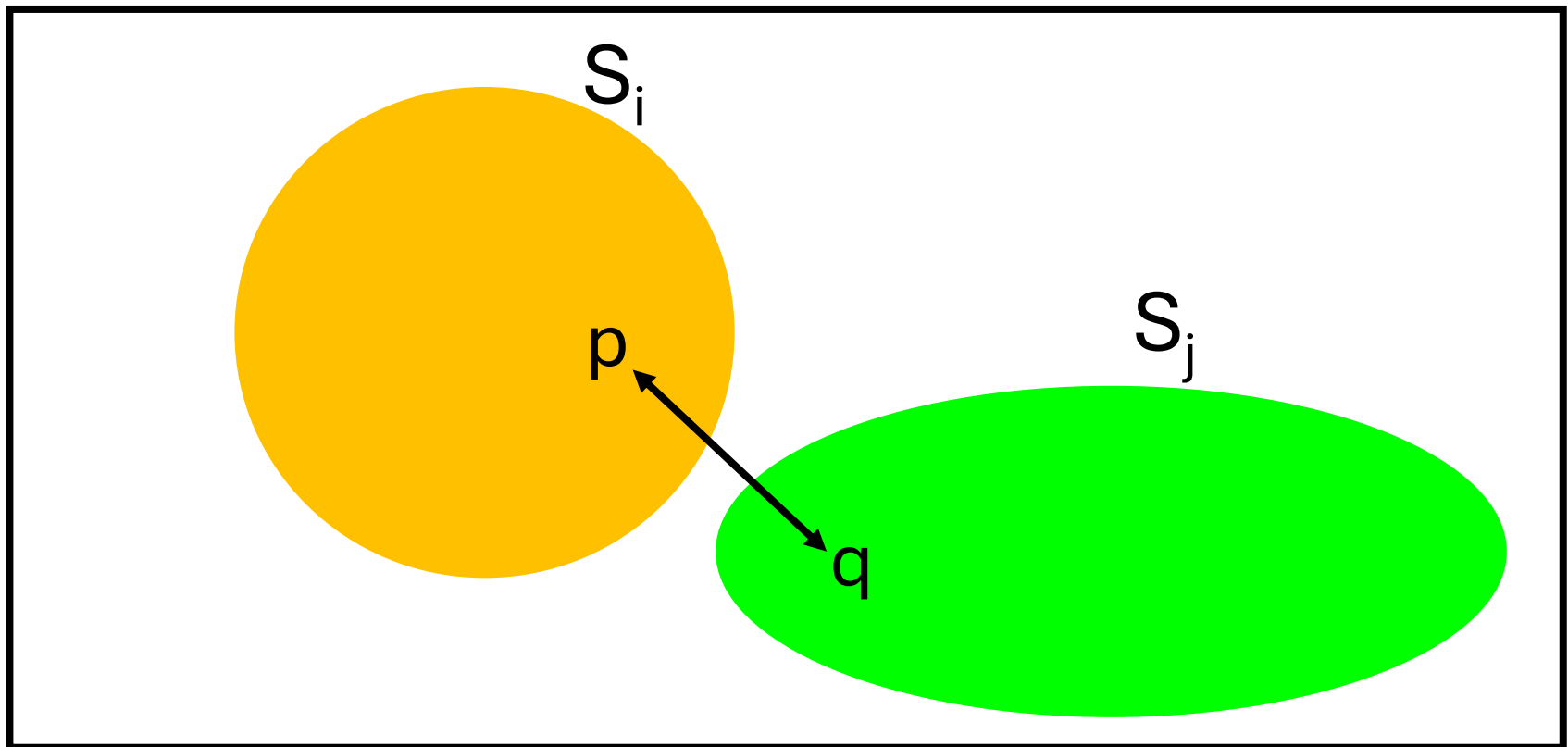
2	1	2
3	3	2
0	5	1



Region

Region

□ **For Two Image Subsets:** S_i and S_j are adjacent if $P \in S_i$ and $q \in S_j$ such that p and q are adjacent.



Region

- Let R be a subset of pixels in an image, two regions R_i and R_j are said to be adjacent if their union form a connected set
- Regions that are not connected are said to be disjoint
- We consider 4- and 8- adjacency when referring to the regions.

Example: $V = \{1\}$

R_i		
1	1	0
1	0	1
1	1	0

R_j		
0	1	1
0	1	1
1	1	1

Numerical Examples

Ques11: Consider the two image subsets, S1 and S2, shown in the following figure. For $V=\{1\}$, determine whether these two subsets are a) 4-adjacent, b) 8-adjacent, or c) m-adjacent.

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	S1					S2				
0	0	0	0	0	0	0	0	1	1	0
1	0	0	1	0	0	0	1	0	0	1
1	0	0	1	0	0	1	1	0	0	0
0	0	1	1	1	0	0	0	0	0	0
0	0	1	1	1	0	0	1	1	1	1

Distance Measure

Distance Measure

□ The following are the different distance measures

- Euclidean Distance: $D_e(p, q) = [(u-w)^2 + (v-x)^2]^{1/2}$
- City Block Distance: $D_4(p, q) = |u-w| + |v-x|$
- Chess Board Distance: $D_8(p, q) = \max(|u-w|, |v-x|)$

Distance Measure

- Given pixels $p(u, v)$, $q(w, x)$ and $r(y, z)$ with coordinates, the distance function D has following properties:
- $D(p, q) \geq 0$ [$D(p, q) = 0$, iff $p = q$]
 - $D(p, q) = D(q, p)$
 - $D(p, r) \leq D(p, q) + D(q, r)$

Euclidean Distance

- For distance measure, the pixels having a distance less than or equal to some value from $p(u, v)$ are the points contained in a disk of radius (r) centered at $p(u, v)$

$$D_e(p, q) = [(u - w)^2 + (v - x)^2]^{1/2}$$

De =

4.2426	3.6056	3.1623	3.0000	3.1623	3.6056	4.2426
3.6056	2.8284	2.2361	2.0000	2.2361	2.8284	3.6056
3.1623	2.2361	1.4142	1.0000	1.4142	2.2361	3.1623
3.0000	2.0000	1.0000	0	1.0000	2.0000	3.0000
3.1623	2.2361	1.4142	1.0000	1.4142	2.2361	3.1623
3.6056	2.8284	2.2361	2.0000	2.2361	2.8284	3.6056
4.2426	3.6056	3.1623	3.0000	3.1623	3.6056	4.2426

City-Block Distance

- The pixels having a D_4 distance from $p(u, v)$ less than or equal to some value, form a diamond centered at $p(u, v)$
- The pixels with $D_4 = 1$ are the 4-neighbors of $p(u, v)$

$$D_4(p, q) = |u - w| + |v - x|$$

D4 =

6	5	4	3	4	5	6
5	4	3	2	3	4	5
4	3	2	1	2	3	4
3	2	1	0	1	2	3
4	3	2	1	2	3	4
5	4	3	2	3	4	5
6	5	4	3	4	5	6

Chessboard Distance

- The pixels with D_8 distance from $p(u, v)$ less than or equal to some value, form a square centered at $p(u, v)$
- The pixels with $D_8 = 1$ are the 8-neighbors of (u, v)

$$D_8(p, q) = \max(|u - w|, |v - x|)$$

$D_8 =$

3	3	3	3	3	3	3
3	2	2	2	2	2	3
3	2	1	1	1	2	3
3	2	1	0	1	2	3
3	2	1	1	1	2	3
3	2	2	2	2	2	3
3	3	3	3	3	3	3

Numerical Examples

Ques12: Let $V=\{0,1\}$. Compute the D_e , D_4 , and D_8 distances between two pixels p and q . Find the distance measures.

	0	1	2	3
0	0	1	1	1
1	1	0	0	1
2	1	1	1	1(q)
3	1(p)	1	1	1

Numerical Examples

Ques13: An image segment is shown below. Let, $V = \{2, 3\}$ be the set of gray level values used to define connectivity in the image. Compute D_e , D_4 , and D_8 distances between pixels 'p' and 'q'.

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2 (p)	3	2	6	1
6	2	3	6	2
5	3	2	3	5
2	4	3	5	2
4	5	2	3	6 (q)

Numerical Examples

Ques14: Consider the following 1-bit image

P: 1	1	1	1
1	1	0	1
0	0	0	1
0	0	1	0
0	0	1	Q:1

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- Calculate the city block distance between P and Q.
- Draw the m-adjacency path from Pixel P to Pixel Q where $V = \{1\}$
- Find the size of the image.

Numerical Examples

Ques15: Consider the following 1-bit image

P: 1	1	1	0	0
1	1	0	0	1
1	0	1	1	1
1	1	1	0	Q:1
1	0	1	0	0

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- Calculate the city block distance between P and Q.
- Draw the m-adjacency path from Pixel P to Pixel Q where $V = \{1\}$
- What will be the size of the given image if gray level range is [0 to 15]

Numerical Examples

Ques16: In two separate 7×7 grids, mark all the pixels which are at a distance of ≤ 3 from pixel located at (3,3), when the distance is (i) City-block (ii) Chess-board

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Numerical Examples

Ques17: The distance measure $D(p, q) = |a-r| + |b-c|$, where the coordinates of point p are (a, b) and that of q are (r, c) is known as

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Thank You